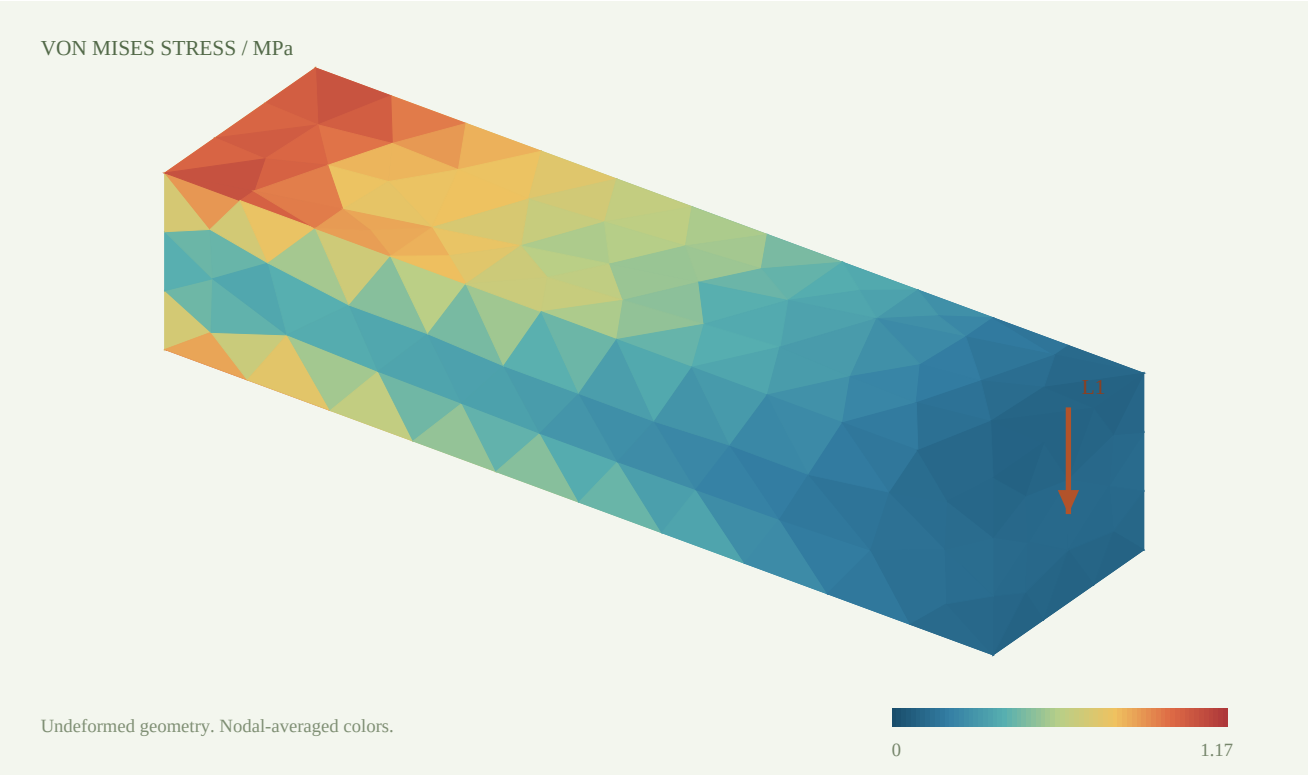
print-coupon

Run 12f9942450724ed1876b14254905a847 | 2026-09-22T06:46:38Z

PEAK VON MISES	MAX DISPLACEMENT	YIELD RATIO
1.461 MPa	0.09957 mm	27.37



Interpretation

This is a linear-elastic engineering estimate. Peak stress is the maximum unaveraged quadrature-point von Mises stress. The diagram uses volume-weighted nodal averages on undeformed geometry.

What to trust

- Maximum displacement 0.09957 mm is the number to compare when you change print settings or loads.
- The solve residual is 4.44e-10. Below 1e-7 means the algebra finished; it does not mean the mesh or material is right.
- Estimated filament mass 2.517 g (63.4% solid equivalent) tracks walls, skins and infill. It is not a weighed print.

Do not treat as printed strength

- Peak von Mises is the highest unaveraged integration-point stress. Fixture edges and sharp corners are often singular — that peak will not settle with a finer mesh.
- Yield ratio uses that peak and the supplied yield. It is not a safety factor and not printed-part strength.
- The print model is a homogenized equivalent solid, not a sliced toolpath and not a calibration against printed coupons.

Pin this run, change infill, walls or orientation, and solve again. Keep other conditions fixed and check mesh convergence before interpreting trends.

A single mesh does not establish convergence or physical acceptance. Yield ratio uses the supplied material strength and is not a safety certification.

Force equilibrium error: 4.720e-11 N. Free-DOF relative residual: 4.442e-10.

Simulation definition

Model	print-coupon
Analysis	3D small-strain static linear elasticity
Compute device	apple · metal-3
Solver method	WebGPU conjugate gradient, symmetric Jacobi scaling
Precision	GPU float32 with browser float64 iterative refinement
Setup proposal	Manually assigned
Material	PLA · illustrative
Young modulus	3500 MPa
Poisson ratio	0.35
Supplied yield strength	40 MPa
Density	1240 kg/m3
Geometry source	stl_import
Solid volume	3200 mm3
Mesh volume	3200 mm3
Estimated mass	2.5168 g
Mesh	807 tetrahedra / 4,455 quadratic DOFs
Print model	Homogenized FDM equivalent solid (not sliced toolpaths)
Print volume method	Actual layer contours
Infill	gyroid at 35%
Walls / skins	3 walls × 0.4 mm; 4+4 skins × 0.2 mm
Print rotation	[30.0, 45.0, 20.0] deg (print Z = layer normal)
Effective E in-plane / layer	2095.87 / 2026.94 MPa
Filament volume / mass	2029.7 mm3 / 2.5168 g
Solid volume fraction	0.6343
Nominal mesh size	7 mm
Minimum element quality	0.32558 (minSICN)
Fixed faces	9 (XYZ translations = 0)

Boundary conditions and verification

L1: face 7, total XYZ force [0.0, 0.0, -4.0] N, distributed uniformly over 80 mm2.

Support reaction XYZ: [4.451772284141953e-11, -1.5580321754971038e-11, 4.0000000000019] N. Moment equilibrium error: 5.641e-10 N mm.

Strain energy: 0.19686 N mm.

Application notes

Preserved setup notes. The print volume method in the simulation definition identifies this run's calculation.

Use the unchanged built 40 × 10 × 8 mm imported coupon. Face 9 at X=0 is fully restrained in XYZ. Face 7 at X=40 receives TOTAL force [0,0,-4] N uniformly distributed over its area; all other surfaces are free. Replace prior bracket application notes and conditions. Print rotation is extrinsic X then Y then Z, R=Rz Ry Rx, mapping print axes into global axes; print Z is the layer normal. Force remains in global axes. Experimental directional homogenization uses illustrative pattern coefficients and bounding-box wall/skin estimates, not sliced infill or calibrated printed strength. No results or physical acceptance are asserted.

Astra setup assumptions:
- User explicitly adopts illustrative PLA properties and all listed print settings for software verification.
- Steady room-temperature, small-strain static loading.
- No self-weight, contact, bolts or preload.
- Print orientation and physical strength are not validated by this approximate model.

Assumptions and provenance

- Printed equivalent solid with directional homogenized stiffness; tetrahedral small-strain static analysis.

- Fixtures are ideal fully fixed faces. Forces are distributed uniformly across the selected mesh faces in global XYZ.

- No contact, bolt preload, plasticity, buckling, fatigue or thermal effects. Explicit infill geometry is not meshed.

- Material presets are illustrative inputs. Verify properties for the actual alloy, condition and manufacturing process.

- Peak stress is the maximum unaveraged quadrature-point von Mises stress; display colors are volume-weighted nodal averages.

- Sharp corners and fixture edges can have stress singularities. A single mesh does not establish stress convergence.

- Yield ratio is a linear-elastic indicator, not a safety certification. Physical acceptance requires independent validation.

- Geometry is an imported triangle mesh. Selectable surfaces are clustered patches, not CAD faces.

- Displacement interpolation is quadratic (P2).

- Print stiffness is a homogenized equivalent solid, not an explicit infill mesh or sliced toolpath.

- Perimeter and skin volumes use actual layer contours, including internal holes and local top/bottom surfaces.

- Shell and core fractions are homogenized throughout the body; the spatial stiffness of individual walls and infill is not resolved.

- Layering is transversely isotropic: print Z is the weak/layer direction after the supplied rotation.

- gyroid core factors are illustrative (in-plane 0.82, layer 0.72), not coupon-calibrated.

- Filament modulus, strength and density remain user inputs. No printed-specimen calibration is claimed.

- Additional raster-scale anisotropy, local voids, bonding defects and moisture are not modelled.

- Layer contours use nominal line width and layer height; bead overlap, variable extrusion widths, gap fill and toolpaths are not modeled.

Reproducibility

Geometry revision: 9737ceeb435f40c6855ecef6bacf518a. Solver versions: {'scikit-fem': '12.0.2', 'numpy': '2.5.3', 'scipy': '1.18.1', 'gmsh': '4.15.2'}. Runtime: 2.23 s. No mesh-convergence study was performed for this run.

Geometry SHA-256: 1ea8b0211b1061e26c5c8456067b898ca0399bb72aeea3e895438bf2d8fce523

Mesh SHA-256: 36659bf625a74bebb46dc387e6ca03c76eae3e0e8612c3db2267e79c621ca380

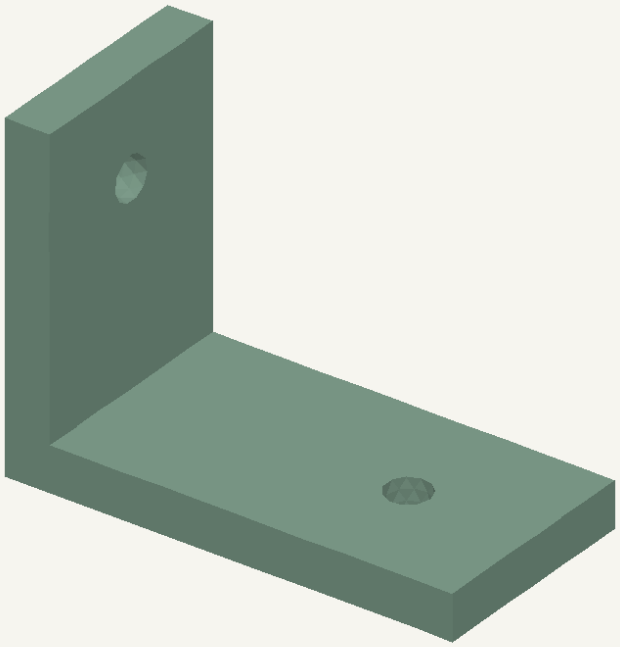
The exported package includes available geometry files, mesh, inputs, results, complete solution arrays, nodal CSV, solver source and pinned dependencies. Material values are supplied inputs, not certified data.

Attached reference context

Preserved with this study; not proof that the simulated geometry was reconstructed from this image. Check the application notes and interview for its relevance.

FEA-R / DEMONSTRATION REFERENCE

Synthetic CAD render. Dimensions and scale intentionally omitted.



Reference 01 / use the annotation tools to describe measured dimensions.

Annotation 1 (dimension): 80 mm overall base length (supplied demo specification)

Normalized image coordinates: [0.23424369747899154, 0.7241115196078431] to [0.6081932773109243, 0.9201899509803922]